

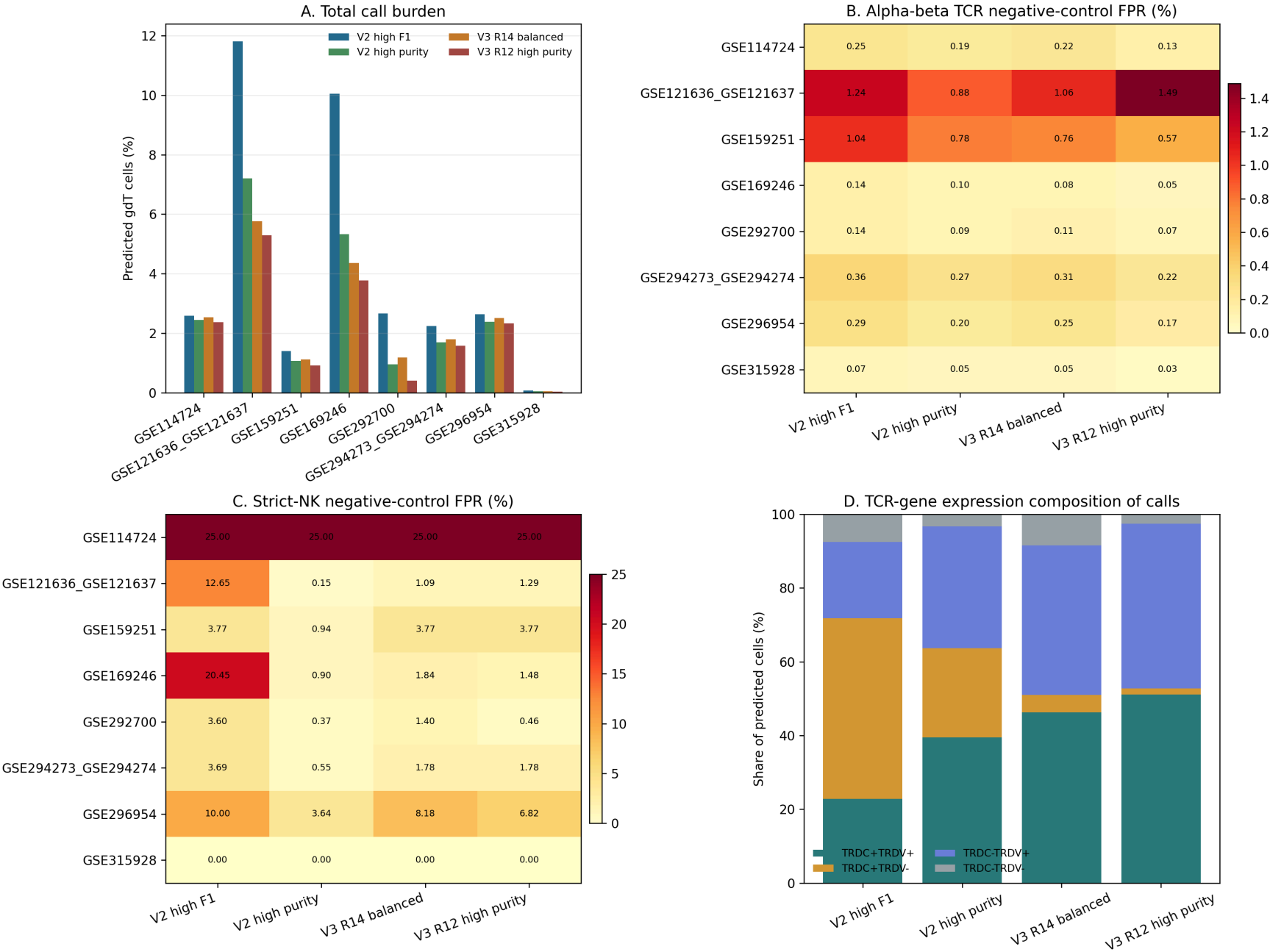
Frozen gdTAI V2/V3 extension-cohort negative-control evaluation

All four registered operating profiles were applied directly to the eight independently TNK-filtered cohorts using `log1p(raw counts per 10,000)` . Model files were checksum-pinned and every H5AD was opened read-only.

Interpretation boundary: these studies used alpha-beta TCR assays and contributed 0 gdT-gold and 0 gdT-silver cells under the project TCR rules. Therefore this report estimates false-positive behavior and prediction burden only. It does not estimate recall, precision, F1, ROC-AUC, or PR-AUC on these cohorts, and it does not promote a model.

758,135 TNK-filtered cells screened	8 extension cohorts
8 represented RNA GSE accessions	v2_high_purity lowest pooled known-negative FPR (0.350%)

Main Results



profile_id	all_calls	all_call_rate	abT_FP	abT_FPR	paired_abT_FP	paired_abT_FPR	single_abT_FP	single_abT_FPR	strict_NK_FP	strict_NK_FPR	known_negative_FP	known_negative_FPR	no_TCR_calls
v2_high_f1	45204	5.963%	1237	0.366%	561	0.199%	676	1.203%	10770	14.869%	12007	2.925%	43967
v2_high_purity	25441	3.356%	912	0.270%	411	0.146%	501	0.891%	525	0.725%	1437	0.350%	24529
v3_round14_balanced	22151	2.922%	962	0.285%	413	0.147%	549	0.977%	1242	1.715%	2204	0.537%	21189
v3_round12_high_purity	18922	2.496%	767	0.227%	299	0.106%	468	0.833%	872	1.204%	1639	0.399%	18155

Trade-off: v3_round12_high_purity has the lowest pooled alpha-beta TCR FPR (0.227%), whereas v2_high_purity has the lowest strict-NK FPR (0.725%). After excluding the reduced-feature GSE169246 sensitivity case, the lowest known-negative union FPR is v3_round12_high_purity (0.295%). Historical positive sensitivity still favors V3 Round 14 over Round 12. These data therefore do not support replacing the current operating profiles with a single new winner.

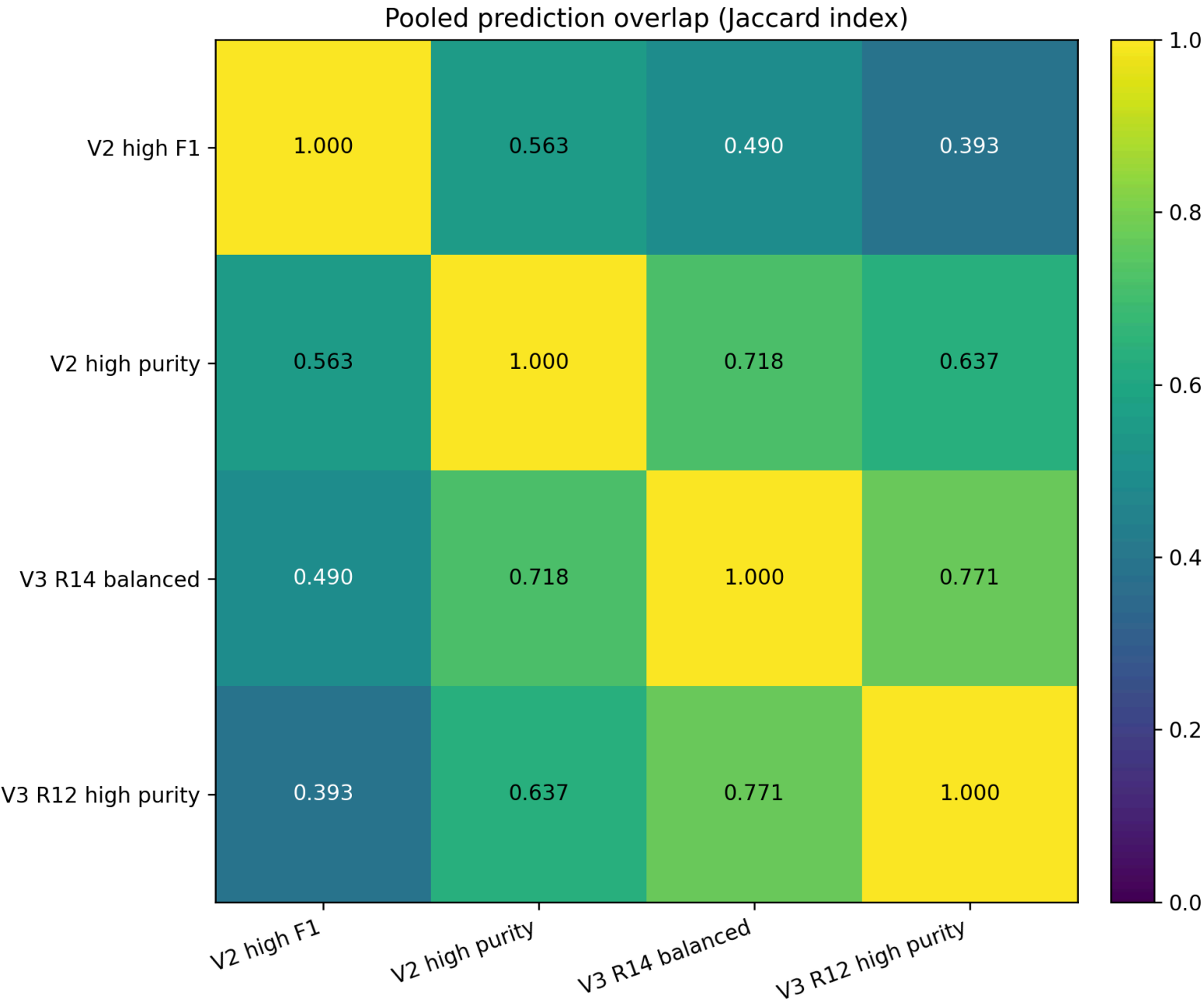
A productive single TRA or single TRB chain is accepted as alpha-beta negative evidence, as requested. Strict NK controls require no productive alpha-beta or gamma-delta chain evidence. Calls among cells with no productive TCR are candidates, not validated true positives.

Feature-coverage sensitivity analysis

scope	n_cohorts	profile_id	stratum	n_cells	predicted_gdT	call_rate	wilson_95_low	wilson_95_high
all_cohorts	8	v2_high_f1	abT_any_chain	338046	1237	0.366%	0.346%	0.387%
all_cohorts	8	v2_high_purity	abT_any_chain	338046	912	0.270%	0.253%	0.288%
all_cohorts	8	v3_round14_balanced	abT_any_chain	338046	962	0.285%	0.267%	0.303%
all_cohorts	8	v3_round12_high_purity	abT_any_chain	338046	767	0.227%	0.211%	0.244%
all_cohorts	8	v2_high_f1	strict_NK	72434	10770	14.869%	14.611%	15.130%
all_cohorts	8	v2_high_purity	strict_NK	72434	525	0.725%	0.666%	0.789%
all_cohorts	8	v3_round14_balanced	strict_NK	72434	1242	1.715%	1.623%	1.812%
all_cohorts	8	v3_round12_high_purity	strict_NK	72434	872	1.204%	1.127%	1.286%
all_cohorts	8	v2_high_f1	known_negative_union	410480	12007	2.925%	2.874%	2.977%
all_cohorts	8	v2_high_purity	known_negative_union	410480	1437	0.350%	0.332%	0.369%
all_cohorts	8	v3_round14_balanced	known_negative_union	410480	2204	0.537%	0.515%	0.560%
all_cohorts	8	v3_round12_high_purity	known_negative_union	410480	1639	0.399%	0.380%	0.419%
complete_schema_sensitivity	7	v2_high_f1	abT_any_chain	283121	1162	0.410%	0.388%	0.435%
complete_schema_sensitivity	7	v2_high_purity	abT_any_chain	283121	856	0.302%	0.283%	0.323%
complete_schema_sensitivity	7	v3_round14_balanced	abT_any_chain	283121	916	0.324%	0.303%	0.345%
complete_schema_sensitivity	7	v3_round12_high_purity	abT_any_chain	283121	737	0.260%	0.242%	0.280%
complete_schema_sensitivity	7	v2_high_f1	strict_NK	25176	1107	4.397%	4.151%	4.657%
complete_schema_sensitivity	7	v2_high_purity	strict_NK	25176	102	0.405%	0.334%	0.492%
complete_schema_sensitivity	7	v3_round14_balanced	strict_NK	25176	371	1.474%	1.332%	1.630%
complete_schema_sensitivity	7	v3_round12_high_purity	strict_NK	25176	171	0.679%	0.585%	0.788%
complete_schema_sensitivity	7	v2_high_f1	known_negative_union	308297	2269	0.736%	0.706%	0.767%
complete_schema_sensitivity	7	v2_high_purity	known_negative_union	308297	958	0.311%	0.292%	0.331%

scope	n_cohorts	profile_id	stratum	n_cells	predicted_gdT	call_rate	wilson_95_low	wilson_95_high
complete_schema_sensitivity	7	v3_round14_balanced	known_negative_union	308297	1287	0.417%	0.395%	0.441%
complete_schema_sensitivity	7	v3_round12_high_purity	known_negative_union	308297	908	0.295%	0.276%	0.314%
schema_warning_cohorts	1	v2_high_f1	abT_any_chain	54925	75	0.137%	0.109%	0.171%
schema_warning_cohorts	1	v2_high_purity	abT_any_chain	54925	56	0.102%	0.079%	0.132%
schema_warning_cohorts	1	v3_round14_balanced	abT_any_chain	54925	46	0.084%	0.063%	0.112%
schema_warning_cohorts	1	v3_round12_high_purity	abT_any_chain	54925	30	0.055%	0.038%	0.078%
schema_warning_cohorts	1	v2_high_f1	strict_NK	47258	9663	20.447%	20.086%	20.813%
schema_warning_cohorts	1	v2_high_purity	strict_NK	47258	423	0.895%	0.814%	0.984%
schema_warning_cohorts	1	v3_round14_balanced	strict_NK	47258	871	1.843%	1.726%	1.968%
schema_warning_cohorts	1	v3_round12_high_purity	strict_NK	47258	701	1.483%	1.378%	1.596%
schema_warning_cohorts	1	v2_high_f1	known_negative_union	102183	9738	9.530%	9.351%	9.712%
schema_warning_cohorts	1	v2_high_purity	known_negative_union	102183	479	0.469%	0.429%	0.513%
schema_warning_cohorts	1	v3_round14_balanced	known_negative_union	102183	917	0.897%	0.841%	0.957%
schema_warning_cohorts	1	v3_round12_high_purity	known_negative_union	102183	731	0.715%	0.666%	0.769%

Agreement Between Profiles



Pooled cell-level Jaccard overlap. Detailed cohort and negative-control discordances are in [screen_pairwise_overlap.csv](#).

Cohort-by-Cohort Audit

dataset_id	profile_id	total_cells	predicted_gdT	predicted_rate	abT_n	abT_fp	abT_fpr	strict_NK_n	strict_NK_fp	strict_NK_fpr
GSE114724	v2_high_f1	28652	742	2.590%	26893	67	0.249%	8	2	25.000%
GSE114724	v2_high_purity	28652	701	2.447%	26893	51	0.190%	8	2	25.000%
GSE114724	v3_round12_high_purity	28652	678	2.366%	26893	36	0.134%	8	2	25.000%
GSE114724	v3_round14_balanced	28652	726	2.534%	26893	58	0.216%	8	2	25.000%
GSE121636_GSE121637	v2_high_f1	18793	2219	11.808%	11439	142	1.241%	2016	255	12.649%
GSE121636_GSE121637	v2_high_purity	18793	1353	7.199%	11439	101	0.883%	2016	3	0.149%
GSE121636_GSE121637	v3_round12_high_purity	18793	995	5.295%	11439	170	1.486%	2016	26	1.290%
GSE121636_GSE121637	v3_round14_balanced	18793	1084	5.768%	11439	121	1.058%	2016	22	1.091%
GSE159251	v2_high_f1	68898	965	1.401%	53773	557	1.036%	106	4	3.774%
GSE159251	v2_high_purity	68898	733	1.064%	53773	420	0.781%	106	1	0.943%
GSE159251	v3_round12_high_purity	68898	627	0.910%	53773	308	0.573%	106	4	3.774%
GSE159251	v3_round14_balanced	68898	768	1.115%	53773	407	0.757%	106	4	3.774%
GSE169246	v2_high_f1	354373	35620	10.052%	54925	75	0.137%	47258	9663	20.447%
GSE169246	v2_high_purity	354373	18869	5.325%	54925	56	0.102%	47258	423	0.895%
GSE169246	v3_round12_high_purity	354373	13384	3.777%	54925	30	0.055%	47258	701	1.483%
GSE169246	v3_round14_balanced	354373	15440	4.357%	54925	46	0.084%	47258	871	1.843%
GSE292700	v2_high_f1	78254	2080	2.658%	24577	34	0.138%	21362	770	3.605%
GSE292700	v2_high_purity	78254	749	0.957%	24577	23	0.094%	21362	80	0.374%
GSE292700	v3_round12_high_purity	78254	312	0.399%	24577	16	0.065%	21362	98	0.459%
GSE292700	v3_round14_balanced	78254	929	1.187%	24577	26	0.106%	21362	299	1.400%
GSE294273_GSE294274	v2_high_f1	55744	1246	2.235%	36221	129	0.356%	1464	54	3.689%
GSE294273_GSE294274	v2_high_purity	55744	943	1.692%	36221	98	0.271%	1464	8	0.546%
GSE294273_GSE294274	v3_round12_high_purity	55744	882	1.582%	36221	81	0.224%	1464	26	1.776%
GSE294273_GSE294274	v3_round14_balanced	55744	998	1.790%	36221	113	0.312%	1464	26	1.776%
GSE296954	v2_high_f1	86608	2283	2.636%	63405	184	0.290%	220	22	10.000%
GSE296954	v2_high_purity	86608	2058	2.376%	63405	128	0.202%	220	8	3.636%
GSE296954	v3_round12_high_purity	86608	2023	2.336%	63405	105	0.166%	220	15	6.818%
GSE296954	v3_round14_balanced	86608	2171	2.507%	63405	156	0.246%	220	18	8.182%
GSE315928	v2_high_f1	66813	49	0.073%	66813	49	0.073%	0	0	
GSE315928	v2_high_purity	66813	35	0.052%	66813	35	0.052%	0	0	
GSE315928	v3_round12_high_purity	66813	21	0.031%	66813	21	0.031%	0	0	
GSE315928	v3_round14_balanced	66813	35	0.052%	66813	35	0.052%	0	0	

Wilson 95% confidence intervals, source-GSE rows, single-chain versus paired-chain strata, and donor/library results are provided in the canonical CSV tables.

Failure-Mode Inspection

TRDC/TRDV expression among known false positives

profile_id	population	tcr_gene_quadrant	n_selected	n_cells	fraction_of_selected
v2_high_f1	abT_false_positives	TRDC+TRDV+	1237	367	29.669%
v2_high_f1	abT_false_positives	TRDC+TRDV-	1237	48	3.880%
v2_high_f1	abT_false_positives	TRDC-TRDV+	1237	592	47.858%
v2_high_f1	abT_false_positives	TRDC-TRDV-	1237	230	18.593%
v2_high_f1	strict_NK_false_positives	TRDC+TRDV+	10770	290	2.693%
v2_high_f1	strict_NK_false_positives	TRDC+TRDV-	10770	9181	85.246%
v2_high_f1	strict_NK_false_positives	TRDC-TRDV+	10770	440	4.085%
v2_high_f1	strict_NK_false_positives	TRDC-TRDV-	10770	859	7.976%
v2_high_purity	abT_false_positives	TRDC+TRDV+	912	287	31.469%
v2_high_purity	abT_false_positives	TRDC+TRDV-	912	25	2.741%
v2_high_purity	abT_false_positives	TRDC-TRDV+	912	469	51.425%
v2_high_purity	abT_false_positives	TRDC-TRDV-	912	131	14.364%
v2_high_purity	strict_NK_false_positives	TRDC+TRDV+	525	217	41.333%
v2_high_purity	strict_NK_false_positives	TRDC+TRDV-	525	138	26.286%
v2_high_purity	strict_NK_false_positives	TRDC-TRDV+	525	151	28.762%
v2_high_purity	strict_NK_false_positives	TRDC-TRDV-	525	19	3.619%
v3_round14_balanced	abT_false_positives	TRDC+TRDV+	962	330	34.304%
v3_round14_balanced	abT_false_positives	TRDC+TRDV-	962	9	0.936%
v3_round14_balanced	abT_false_positives	TRDC-TRDV+	962	502	52.183%
v3_round14_balanced	abT_false_positives	TRDC-TRDV-	962	121	12.578%
v3_round14_balanced	strict_NK_false_positives	TRDC+TRDV+	1242	288	23.188%
v3_round14_balanced	strict_NK_false_positives	TRDC+TRDV-	1242	117	9.420%
v3_round14_balanced	strict_NK_false_positives	TRDC-TRDV+	1242	406	32.689%
v3_round14_balanced	strict_NK_false_positives	TRDC-TRDV-	1242	431	34.702%
v3_round12_high_purity	abT_false_positives	TRDC+TRDV+	767	275	35.854%
v3_round12_high_purity	abT_false_positives	TRDC+TRDV-	767	6	0.782%
v3_round12_high_purity	abT_false_positives	TRDC-TRDV+	767	440	57.366%
v3_round12_high_purity	abT_false_positives	TRDC-TRDV-	767	46	5.997%
v3_round12_high_purity	strict_NK_false_positives	TRDC+TRDV+	872	272	31.193%
v3_round12_high_purity	strict_NK_false_positives	TRDC+TRDV-	872	7	0.803%
v3_round12_high_purity	strict_NK_false_positives	TRDC-TRDV+	872	377	43.234%
v3_round12_high_purity	strict_NK_false_positives	TRDC-TRDV-	872	216	24.771%

Highest-rate libraries with at least 100 controls

dataset_id	source_gse_id	group_value	profile_id	stratum	n_cells	predicted_gdT	call_rate	wilson_95_low	wilson_95_high
GSE121636_GSE121637	GSE121636	GSM3440845_GU0715_T	v2_high_f1	abT_any_chain	2220	119	5.360%	4.498%	6.377%
GSE294273_GSE294274	GSE294273	GSM8901618_HM009	v2_high_f1	abT_any_chain	1989	48	2.413%	1.825%	3.185%
GSE169246	GSE169246	Post_P002_t	v2_high_f1	strict_NK	212	90	42.453%	35.992%	49.182%
GSE169246	GSE169246	Prog_P007_b	v2_high_f1	strict_NK	3173	1260	39.710%	38.021%	41.424%
GSE121636_GSE121637	GSE121636	GSM3440845_GU0715_T	v2_high_purity	abT_any_chain	2220	82	3.694%	2.986%	4.562%
GSE294273_GSE294274	GSE294273	GSM8901618_HM009	v2_high_purity	abT_any_chain	1989	35	1.760%	1.268%	2.437%
GSE169246	GSE169246	Post_P018_t	v2_high_purity	strict_NK	175	17	9.714%	6.154%	15.005%
GSE169246	GSE169246	Post_P025_t	v2_high_purity	strict_NK	251	16	6.375%	3.961%	10.103%
GSE121636_GSE121637	GSE121636	GSM3440845_GU0715_T	v3_round12_high_purity	abT_any_chain	2220	155	6.982%	5.995%	8.118%
GSE294273_GSE294274	GSE294273	GSM8901618_HM009	v3_round12_high_purity	abT_any_chain	1989	29	1.458%	1.017%	2.086%
GSE169246	GSE169246	Post_P018_t	v3_round12_high_purity	strict_NK	175	29	16.571%	11.793%	22.786%
GSE169246	GSE169246	Pre_P023_t	v3_round12_high_purity	strict_NK	106	12	11.321%	6.596%	18.751%
GSE121636_GSE121637	GSE121636	GSM3440845_GU0715_T	v3_round14_balanced	abT_any_chain	2220	102	4.595%	3.799%	5.547%
GSE294273_GSE294274	GSE294273	GSM8901618_HM009	v3_round14_balanced	abT_any_chain	1989	42	2.112%	1.566%	2.842%
GSE169246	GSE169246	Post_P018_t	v3_round14_balanced	strict_NK	175	27	15.429%	10.825%	21.517%
GSE169246	GSE169246	Pre_P023_t	v3_round14_balanced	strict_NK	106	11	10.377%	5.894%	17.632%

Historical Positive/Negative Benchmarks

These values come from the frozen models' prior validation frames and are shown only to retain the sensitivity context missing from the new studies. They are not pooled with the extension results because the cohorts and truth definitions differ.

profile_id	evaluation_frame	precision	recall	specificity	f1	source_file
v2_high_f1	packaged V2 validation	94.784%	87.007%	99.663%	90.729%	Integrated_dataset/models/gdT_prediction_classifier/gdTAI_v2.0/mode_metrics.csv
v2_high_purity	packaged V2 validation	95.148%	86.280%	99.690%	90.497%	Integrated_dataset/models/gdT_prediction_classifier/gdTAI_v2.0/mode_metrics.csv
v3_round12_high_purity	historical full-atlas primary gold	98.226%	75.604%	99.869%	85.443%	Integrated_dataset/tables/gdT_prediction/gdtai_v3_round12_vs_round14/full_atlas_performance.csv
v3_round14_balanced	historical full-atlas primary gold	97.521%	80.559%	99.804%	88.233%	Integrated_dataset/tables/gdT_prediction/gdtai_v3_round12_vs_round14/full_atlas_performance.csv

Model and Input Integrity

Frozen artifacts

profile_id	model_id	mode	threshold	gene_features	artifact_sha256	artifact_path
v2_high_f1	gdtai_v2	high_f1	0.9063586592674255	187	c7f6e7d1d1c0c8de9e7abbaa7c5b70c1070b43c964ccdb94e414bc5278e13c3e	Integrated_dataset/models/gdT_prediction_classifier/gdTAI_v2.0/gdTAI_v2_model.pkl
v2_high_purity	gdtai_v2	high_purity	annotation-specific	187	c7f6e7d1d1c0c8de9e7abbaa7c5b70c1070b43c964ccdb94e414bc5278e13c3e	Integrated_dataset/models/gdT_prediction_classifier/gdTAI_v2.0/gdTAI_v2_model.pkl
v3_round14_balanced	gdtai_v3	fixed_threshold	0.936	210	16dedc0081da9b8487887341232bcf8c9c9403dd3bbd72e04cab43d4cd7b2e09	Integrated_dataset/models/gdT_prediction_classifier/gdTAI_v3.0/gdTAI_v3_model.pkl
v3_round12_high_purity	gdtai_v3_round12	fixed_threshold	0.5	210	7373e79350f7db190c415b376b9763e31652754438ee8c5afd3853beb7b2ebc4	Integrated_dataset/models/gdT_prediction_classifier/gdtai_v3_round12_vs_round14/round12_model.pkl

Expression schema

dataset_id	n_cells_evaluated	matrix_encoding	model_gene_fraction	auxiliary_audit_gene_fraction	model_schema_warning	missing_critical_genes
GSE114724	28652	csr_matrix	100.000%	100.000%	False	
GSE121636_GSE121637	18793	csr_matrix	100.000%	100.000%	False	
GSE159251	68898	csr_matrix	90.476%	100.000%	False	
GSE169246	354373	csr_matrix	61.429%	100.000%	True	
GSE292700	78254	csr_matrix	100.000%	100.000%	False	
GSE294273_GSE294274	55744	csr_matrix	100.000%	100.000%	False	
GSE296954	86608	csr_matrix	100.000%	100.000%	False	
GSE315928	66813	csr_matrix	100.000%	100.000%	False	

A schema warning does not mean critical genes are absent. It means fewer than 80% of the full model feature genes were available. Such cohorts must be interpreted separately because zero-filled missing genes can shift predictions.

Methods and Definitions

Input: cohort-separated Phase 1 TNK H5ADs. **Normalization:** raw integer-like X counts were normalized cell-wise to 10,000 and transformed with log1p in chunks. **Models:** V2 high-F1, V2 annotation-specific high-purity, V3 Round 14 balanced, and V3 Round 12 high-purity were loaded only after SHA256 verification.

Alpha-beta negative: productive TRA or TRB, no productive TRG/TRD, and not marked doublet. Paired and single-chain cells are reported separately. **Strict NK negative:** explicit/derived NK classification with no productive TRA/TRB/TRG/TRD and not marked doublet. **Uncertainty:** Wilson 95% binomial intervals. **Paired comparisons:** exact cell-level McNemar tests are descriptive because cells within donors are correlated.

Decision: No model selected or promoted: the new cohorts do not contain an unbiased gdT positive truth set.